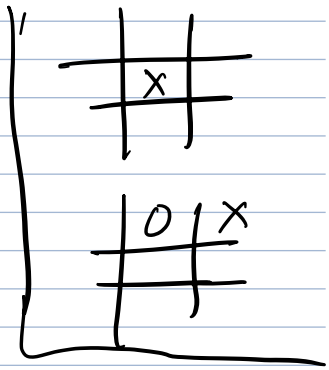
Class 17

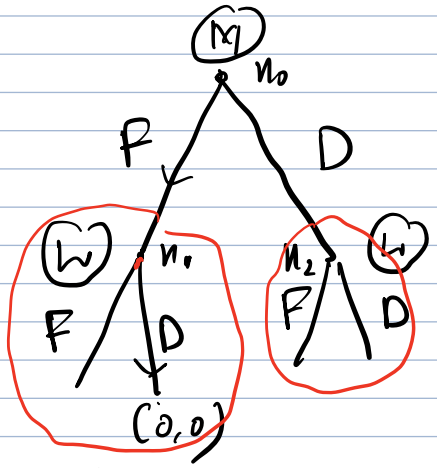
$\langle N, V, E, n_0, (V_i)_{i \in N}, (u_i)_{i \in N} \rangle$

A strategy for player i $s_i : V_i \to \bigcup_{n \in V_i} A(n)$

s.t. $s_i(n) \in A(n)$ $\forall n \in V_i$



- A strategy profile $s = (s_1, \ldots, s_n)$ where s_i is a strategy of player i
- S_i: set of strategies of player i.



$(F, (D,D))$

$u(F, (D,D)) = (0,0)$

$s_2 : V_2 \to \bigcup_{n \in V_2} A(n)$

s.t. $s_2(n) \in A(n)$ $\forall n \in V_2$

$V_2 = \{n_1, n_2\}$

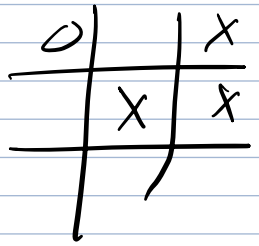
$A(n_1) = \{F, D\}$

$A(n_2) = \{F, D\}$

- Every strategy profile determines a unique path in the game tree, starting from the root and ending at a leaf.

$$(n_0, n_1, \ldots, n_k)$$

$$u(s) := u(n_k)$$



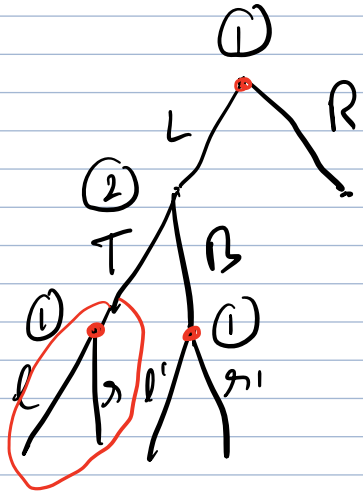
Subgame of a game

Let $\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N}, (u_i)_{i \in N} \rangle$ be an extensive form game. Consider $n \in V$. The the subgame starting at n, denoted by $\Gamma(n)$, is an extensive form game. where

$$\Gamma(n) = \langle N, V(n), E(n), n, (V_i(n))_{i \in N}, (u_i)_{i \in N} \rangle$$

- N = set of players.
- $V(n)$: the set of all nodes that are connected from n.
- $E(n)$: all the edges that connect nodes in $V(n)$
- n the new root.
- $V_i(n)$: $V_i \cap V(n)$
- u_i: the restriction of u_i to the leaves in the game tree $(V(n), E(n), n)$

- $\Gamma = \Gamma(n_0)$
- Also, every leaf defines a subgame where no player makes a move.



$((R, l, l'), T)$

(l, ϕ)

A particular type of games

- Two players.
- three possibilities: (i) P-1 wins $(1,-1)$ (ii) P-2 wins $(-1,1)$ (iii) ends in a draw. $(0,0)$

- A winning strategy for P-1 is a strategy s_1 s.t. $u(s_1, s_2) = (1,-1) \quad \forall s_2 \in S_2$

- A strategy guaranteeing at a draw for P-1 is a strategy s_1 s.t.

$$u(s_1, s_2) \in \{(1,-1), (0,0)\} \quad \forall s_2 \in S_2$$

Similarly, a winning strategy and a strategy

guaranteeing at least a draw can be defined for P-2.

Theorem (Zermelo (1913), Von Neumann (1928))

In every two players game (with perfect information) in which the set of utilities is $\{(1,-1), (-1,1), (0,0)\}$, one and only one of the following holds:

1) P-1 has a winning strategy

2) P-2 has a winning strategy

3) Both players have strategies guaranteeing at least a draw.

Chomp: David Gale (1974)

